

## Interacting Multipoles of Rank 2 and 1:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}\mu_{\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \Theta_{\alpha\beta} \cdot \mu_{\gamma} \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \mu_{\gamma} \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta} \cdot \mu_{\gamma} \\ +3 \cdot R_z^2 \cdot R_{\gamma} \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta} \cdot \mu_{\gamma} \end{bmatrix} \quad (1)$$

## Simplify deltas:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}\mu_{\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \Theta_{\alpha\beta} \cdot \mu_{\gamma} \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \mu_{\beta} \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \mu_{\alpha} \\ +3 \cdot R_z^2 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha} \cdot \mu_{\gamma} \end{bmatrix} \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \mu_{\gamma} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \mu_{\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \mu_{\alpha} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^2 \cdot R_{\gamma} \cdot \mu_{\gamma} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (2b)$$

## Simplify R:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}\mu_{\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Theta_{zz} \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_{\beta} \cdot \Theta_{z\beta} \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_{\alpha} \cdot \Theta_{z\alpha} \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_z \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +3 \cdot R_z^3 \cdot \mu_{\beta} \cdot \Theta_{z\beta} \\ +3 \cdot R_z^3 \cdot \mu_{\alpha} \cdot \Theta_{z\alpha} \end{bmatrix} \quad (3b)$$

## Simplify Dipoles:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}\mu_{\gamma} = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +3 \cdot R_z^3 \cdot \mu_y \cdot \Theta_{yz} \\ +3 \cdot R_z^3 \cdot \mu_y \cdot \Theta_{yz} \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot [+0] \quad (4b)$$

## Combining everything:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}\mu_{\gamma} = +0 \quad (5)$$